

FleetFlow

Transport Operations Management System

1. Project Description

FleetFlow is a web-based transport operations management system designed to centralize the day-to-day activities of a fleet. It enables authorized users to manage vehicles and drivers, create and dispatch trips, record maintenance activities, track fuel usage and expenses, and monitor operational information through a dashboard. The system follows a complete operational workflow from authentication and resource assignment to trip completion and post-trip record management, with validation rules to prevent invalid assignments and inconsistent fleet states.

2. Proposed Functional Areas

Module	Purpose
Authentication & Authorization	Secure login with protected routes and role-based access.
Fleet & Driver Management	Maintain vehicle and driver records, availability, status, and licence information.
Trip Management	Create, dispatch, complete, and cancel trips while enforcing operational constraints.
Maintenance Management	Record maintenance activities and update vehicle availability accordingly.
Fuel & Expense Tracking	Maintain fuel logs and operational expense records with input validation.
Dashboard & Metrics	Present operational KPIs and update them as fleet records change.

3. Suitable SDLC Model

Agile Software Development Life Cycle (Agile SDLC)

Agile is the most suitable model for FleetFlow because the project is modular, developed by a team, and likely to evolve as requirements, validation rules, and user-interface needs are refined. The implementation naturally supports incremental delivery: authentication, fleet management, trip management, maintenance, financial records, and dashboard functionality can be developed and integrated in short iterations.

Why Agile Fits This Project

- Incremental development: each major module can be implemented, tested, and integrated independently.
- Frequent feedback: the team can review the working system after every iteration and adjust requirements quickly.
- Early testing and integration: business rules such as vehicle availability, driver eligibility, and cargo capacity can be validated as soon as the related module is built.
- Parallel teamwork: different members can work on authentication, fleet records, trips, and supporting modules concurrently while integrating through defined APIs.
- Suitable for hackathon-style development: the model prioritizes a working core product first and allows optional features to be added later.